Name: Krishna Raviraj Shimpi
PRN No: 202501040036

5.2.1. Titanic Dataset

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Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named titanic.csv and has been loaded using the pandas library. It contains the following columns:

- Pclass: Passenger class (1 = First, 2 = Second, 3 = Third).
- Gender: Gender of the passenger (male/female).
- Age: Age of the passenger.
- Survived: Survival status (0 = Did not survive, 1 = Survived).
- Fare: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution) :

- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).

- Use the color skyblue for the bars.
- Title the plot as **"Passenger Class Distribution"** .
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (Gender Distribution) :

- Create a pie chart to display the distribution of male and female passengers.
- Use lightblue for males and lightcoral for females.
- Include percentages on the slices (use autopct='%1.1f%%').
- Title the plot as **"Gender Distribution"** .

Histogram (Age Distribution) :

- Create a histogram to visualize the distribution of passengers' ages.
- Use lightgreen for the bars with black edges (edgecolor = 'black').
- Set the number of bins to **8** for the histogram.
- Title the plot as **"Age Distribution"** .
- Label the x-axis as **"Age"** and the y-axis as **"Frequency"** .

Bar Plot (Survival Count) :

- Create a bar plot to show the count of passengers who survived and those who did not, based on the Survived column.
- Use the colors lightblue for survivors (1) and lightcoral for non-survivors (0).
- Title the plot as **"Survival Count"** .
- Label the x-axis as **"Survived (0 = No, 1 = Yes)"** and the y-axis as **"Count"**.

Scatter Plot (Fare vs Age) :

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.
- Use orange for the data points.
- Title the plot as **"Fare vs Age"**.
- Label the x-axis as **"Age"** and the y-axis as **"Fare"**.

Note: Refer to the displayed plot in the sample test cases for better understanding.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))
# write the code..
axes[0,0].bar(df['Pclass'].value_counts().index,df['Pclass'].value_counts(),color='skyblue
')
axes[0,0].set_title('Passenger Class Distribution')
axes[0,0].set_xlabel('Pclass')
axes[0,0].set_ylabel('Count')
gender_counts=df['Gender'].value_counts()
axes[0,1].pie(gender_counts,labels=gender_counts.index,colors=['lightblue','lightcoral'])
axes[0,1].set_title('Gender Distribution')
axes[1,0].hist(df['Age'].dropna(),bins=8,color='lightgreen',edgecolor='black')
axes[1,0].set_title('Age Distribution')
axes[1,0].set_xlabel('Age')
axes[1,0].set_ylabel('Frequency')
survival_counts=df['Survived'].value_counts()
axes[1,1].bar(survival_counts.index,survival_counts,color=['lightblue','lightcoral'])
axes[1,1].set_title('Survival Count')
axes[1,1].set_xlabel('Survived (0 = No, 1 = Yes)')
axes[1,1].set_ylabel('Count')
axes[2,0].scatter(df['Age'],df['Fare'],color='orange')
```

```
axes[2,0].set_title('Fare vs Age')
axes[2,0].set_xlabel('Age')
axes[2,0].set_ylabel('Fare')
axes[2,1].plot()
axes[2,1].set_xlim(0,1)
axes[2,1].set_ylim(0,1)
plt.tight_layout()
plt.show()
```